Menel BEREKSI

WATER AND ENVIRONMENTAL ENGINEER

Hydrogeology / Environmental geochemistry



Gif-sur-Yvette, France | +33 6 56 66 97 89 | menel.bereksi@gmail.com | Driving licence: B | LinkedIn

Groundwater | Fieldwork and analysis | National and international mobility

PROFILE

Hydrology and Hydrogeology Master's graduate from Université Paris-Saclay, combining field investigations, geochemical experiments and QGIS data analysis. My research addresses groundwater quality and contaminant transport. Seeking a junior role supporting the characterisation, management and protection of water resources.

RESEARCH EXPERIENCE

GEOPS – Université Paris-Saclay

Jan. – June 2025

Master's research internship – Environmental geochemistry

- **Post-mining contaminants.** Investigated arsenic and iron mobility in unconsolidated mine waste and hardpans at Enguialès, focusing on porosity, fluid composition and solid–solution interactions.
- **Experimental campaigns.** Tested **3 mine-waste samples** over **6 rainfall cycles**; conducted water-table fluctuation and hardpan leaching protocols over **5 cycles each**.
- **Geochemical analyses.** Applied **3 batch tests**: EN 12457-2 and NAG in duplicate, TCLP in triplicate; used pH/conductivity measurements, ion chromatography and AAS (Fe/As), plus porosity and CEC characterisation.
- **Interpretation and reporting.** Interpreted SEM images and existing XRD results, processed time series and synthesised findings; oral and poster presentations and round-table participation.

GEOPS – Université Paris-Saclay

May – June 2023

Research internship – Hydrogeology and water quality

- Used **ADES and InfoTerre** data, **QGIS** mapping and nitrate time series to investigate the Beauce multilayer aquifer and the role of low-permeability formations in groundwater protection.

SONATRACH – Research and Development Centre

2020

Master's research internship – Palynology, Boumerdès, Algeria

- Catalogued Upper Miocene palynomorphs from the Lower Chelif Basin using optical microscopy; analysed botanical affinities and contributed to palaeoenvironmental reconstruction.

FIELD INVESTIGATIONS – UNIVERSITY TRAINING

Université Paris-Saclay 2023–2025

- **Aquifer hydrodynamics.** Investigated the Maisse multilayer aquifer through groundwater level measurements, pumping tests in **3 boreholes**, hydrodynamic parameter estimation and analysis of water transfers.
- **Water quality and contamination.** Studied the Crau alluvial aquifer affected by landfill leachate; sampled mine waste and contributed to site instrumentation at Enguialès.
- **Catchment hydrology.** Studied the Hallue catchment: piezometric mapping, sampling, in-situ physicochemical measurements and streamflow measurements; conducted Porchet infiltration tests.

TECHNICAL SKILLS

Hydrogeology and field methods

Groundwater level monitoring and piezometric mapping, pumping tests, hydraulic conductivity; water and soil sampling, physicochemistry, water isotopes.

Geochemistry and laboratory methods

Batch and column leaching, ion chromatography, AAS; porosity, cation exchange capacity, SEM/XRD data interpretation, optical microscopy.

Data analysis and software

QGIS, ADES, InfoTerre; Python, Microsoft Office, ModelMuse, PHREEQC, LaTeX.

Applied geophysics

Electrical resistivity tomography (ERT), ground-penetrating radar (GPR), seismic surveying – university fieldwork.

SCIENTIFIC COMMUNICATIONS

2 presentations in 2025: GS GCEP oral (5 June), IPSL poster (12 June).

Co-author: EGU 2026 abstract, XRF / XANES.

EDUCATION

Master's degree (M2) in Hydrology and Hydrogeology

Université Paris-Saclay, France 2025

Studies: 2022–2025. Quantitative hydrogeology, geochemistry, modelling; contaminated land module.

Master's degree (M2) in Georesources

University of Tlemcen, Algeria 2020

Bachelor's degree in Geology

University of Tlemcen, Algeria 2018

LANGUAGES AND ACTIVITIES

French C2 | English C1 | Arabic C2

Les Amis du Malade – Co-founder, President

2022: recruited volunteer doctors and organised medical interventions.

PAUWES – Project Assistant

Sept.–Dec. 2021: mapped African entrepreneurial ecosystems and coordinated scientific and educational events.